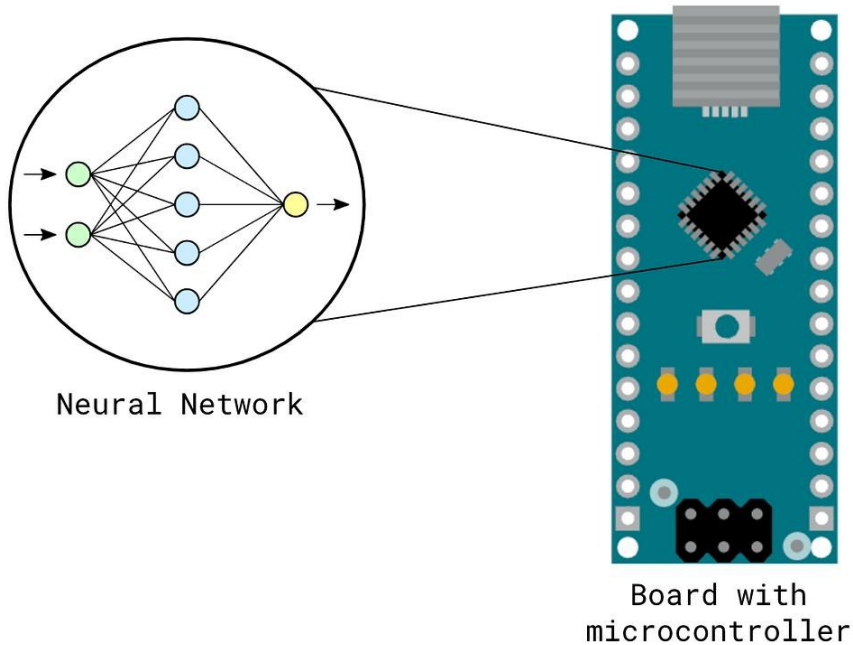


What is TinyML?



Microcontrollers



Wearables



Embedded systems

TinyML = Machine Learning on microcontrollers

- deploying AI models on **ultra-low-power, resource-constrained devices**, such as microcontrollers and digital signal processors
- Runs on KB-level memory, Ultra-low power consumption

Key Characteristics

On-Device Inference.

Minimal Power consumption.

Models optimized for limited storage and RAM.

Cloud AI Vs Edge AI Vs Tiny AI

Cloud AI

- AI models runs on centralized, remote servers (e.g., AWS, Azure)
- **Best for:** Massive data processing, training complex models, and scenarios where latency is not critical.
- **Pros:** High computing power, high scalability, & massive storage.
- **Cons:** Higher latency, requires a constant connection, and potential privacy issues due to data transfer.

Edge AI

- AI model runs —on local devices like smart cameras, autonomous vehicles, or gateways—rather than in a centralized cloud.
- **Best for:** Real-time analytics, privacy-sensitive tasks, and applications with spotty connectivity.
- **Pros:** : Low latency, high privacy, and reduced bandwidth usage.
- **Cons:** Limited processing power and memory of the local device.

Tiny AI (TinyML)

- Subfield of Edge AI - runs ML models on very low-power, constrained hardware, such as MCUs. Operates on a milliwatt power budget.
- **Best for:** Battery-powered (wearables, sensors & small devices).
- **Pros:** Extremely low power consumption (operates on batteries for months/years), very small footprint, and low cost.
- **Cons:** Limited model complexity; highly restricted memory & compute

Cloud data centre

- **Cloud ML**
- DNN on the cloud
- HW, TPU, FP5A, GPU, CPU

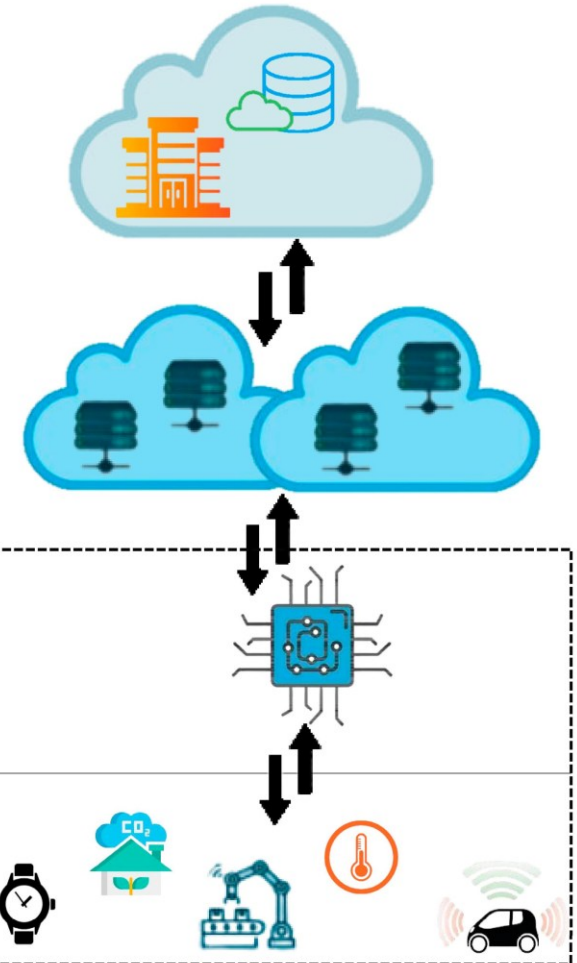
Edge computing

- **EdgeML**
- Optimize alogs and CNN
- Light
- SoC(With NPU/NSP Aacceleator)

Embedded devices

Tiny ML

- CNN micro
- MCU w/HW acceleators



Edge AI vs Tiny ML

Feature	Edge AI	TinyML
Hardware	More powerful: Smartphones, Gateways, Edge Servers, GPUs	Ultra-constrained: Microcontrollers, low- power IoT sensors
Memory	Megabytes (MB) to Gigabytes (GB) of RAM	Kilobytes (KB) of RAM (often < 1MB)
Power Use	Watts (usually requires a plug or large battery)	Milliwatts (can run for years on battery)
Complexity	Can handle high-res video analytics and complex natural language processing	Focuses on simple, "always-on" tasks like wake-word detection or vibration analysis

Edge AI is the broad umbrella for any local processing. A self-driving car processing video feeds is using Edge AI.

TinyML is the "extreme" end of this spectrum, shrinking models to fit into the tiny computers (microcontrollers) found inside everyday objects like toothbrushes or smoke detectors.

Energy Philosophy: Edge AI aims to reduce latency and bandwidth by staying close to the data.

TinyML focuses primarily on **energy autonomy**—enabling intelligence in places where there is no power outlet or reliable internet, such as a sensor in a remote farm field.

Operating Systems: Edge AI devices often run full operating systems like Linux.

TinyML devices frequently run "bare metal" or on very basic Real-Time Operating Systems (RTOS) because they don't have enough memory for a standard OS.

Why TinyML?



Ultra-low power

TinyML is driven by the need to make devices smarter while keeping them efficient.



Works offline

TinyML enables a wide range of applications—from voice recognition to industrial monitoring.



Instant response



**KEYWORD
SPOTTING**



**GESTURE
RECOGNITION**



**ANOMALY
DETECTION**



**HEALTH
MONITORING**

These systems can run for months or even years on small batteries, making them ideal for real-world deployments.



Low-cost devices



High privacy

How AI Works Across Layers

In many modern systems, Cloud AI, **TinyML** and **Edge AI** are complementary. For example, **Smart Voice Assistants (Alexa / Google Assistant)**

Tiny AI (On Device)

- Wake word detection ("Alexa", "Hey Google")
- Always listening, ultra-low power

Edge AI (Device / Gateway)

- Speech-to-text processing
- Noise reduction & intent preprocessing

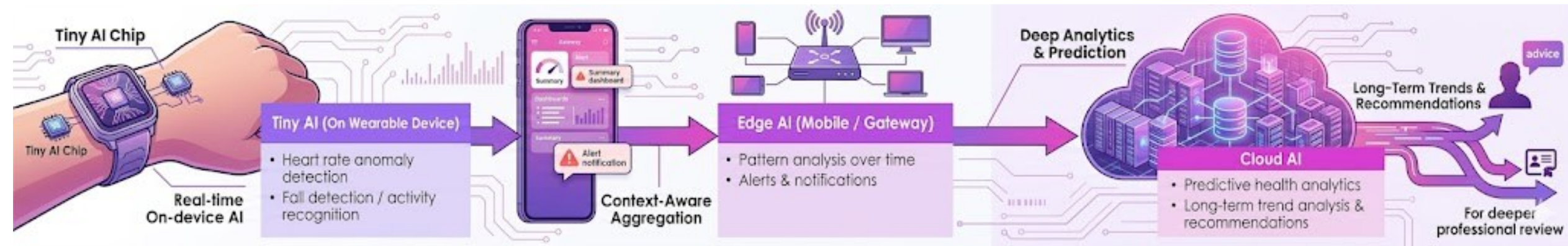
Cloud AI

- Advanced NLP & context understanding
- Continuous learning & personalization



How AI Works Across Layers

Smart Healthcare Wearables



Tiny AI (On Wearable Device)

- Heart rate anomaly detection
- Fall detection / activity recognition

Edge AI (Mobile / Gateway)

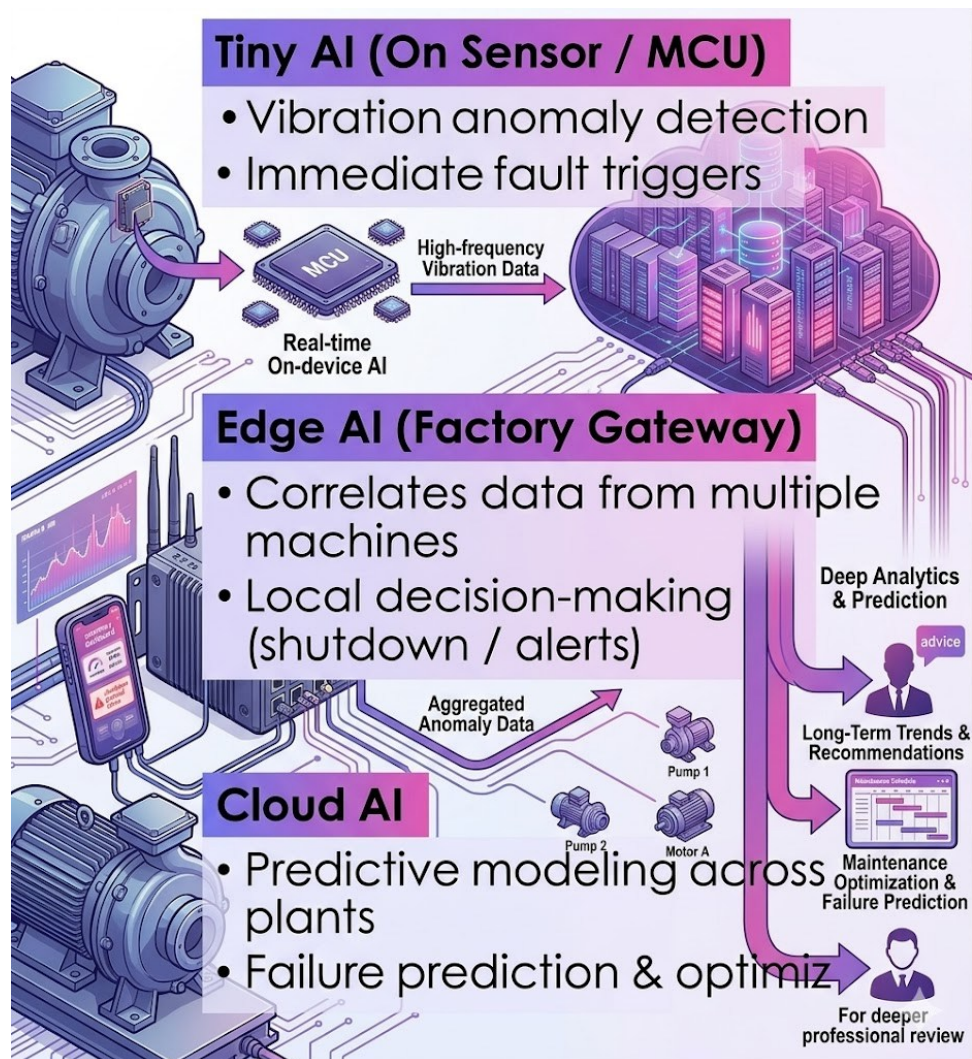
- Pattern analysis over time
- Alerts & notifications

Cloud AI

- Predictive health analytics
- Long-term trend analysis & recommendations

How AI Works Across Layers

Industrial Predictive Maintenance



Tiny AI (On Sensor / MCU)



- constantly monitors high-frequency vibration data. Because this processing happens right on the sensor, the system can detect subtle vibration anomalies and trigger immediate fault responses in real-time without any lag

Edge AI (Factory Gateway)



- gateway aggregates and correlates the anomaly data coming from multiple machines, such as various pumps and motors across the floor leveraging Edge AI

Cloud AI



- all of this refined data is funneled into the Cloud AI for deep, large-scale analytics. the cloud builds complex predictive models that forecast long-term trends and suggest maintenance optimizations

TinyML Pipeline

